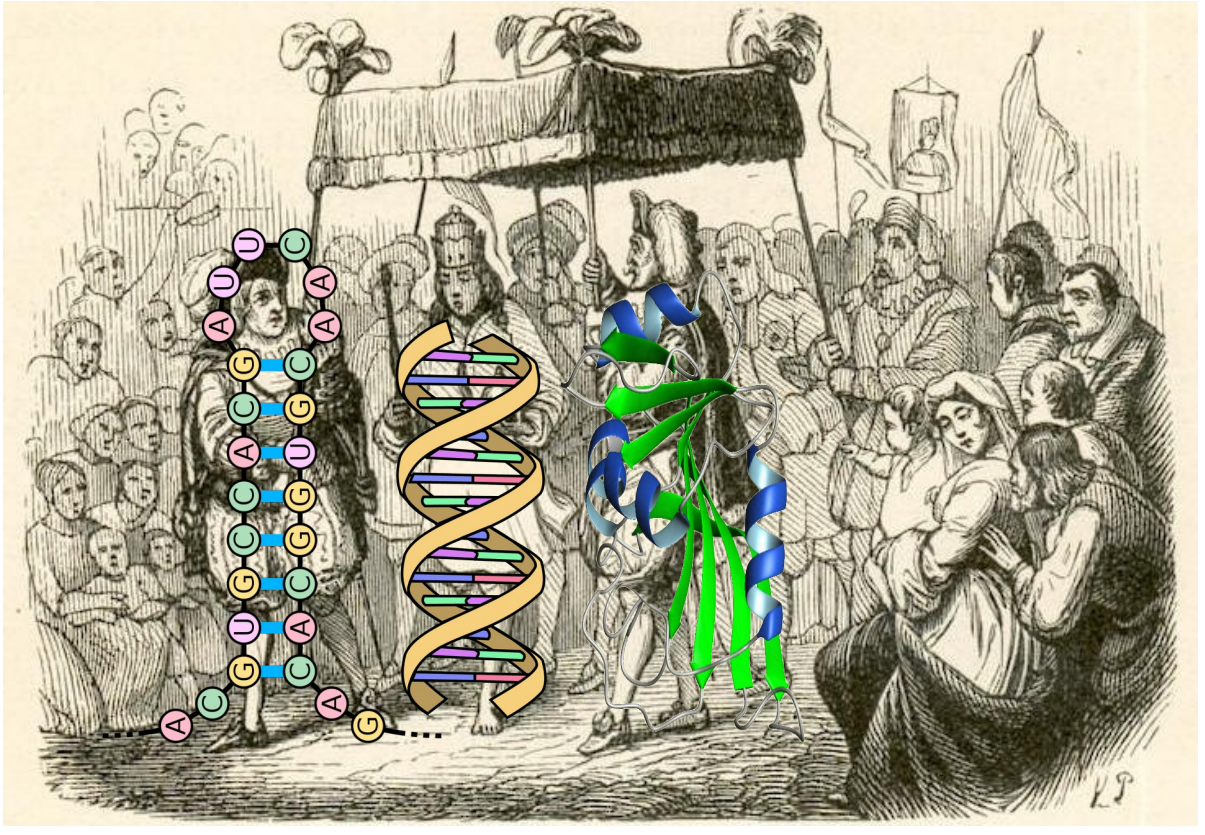
Challenges in Genomics Data Analysis

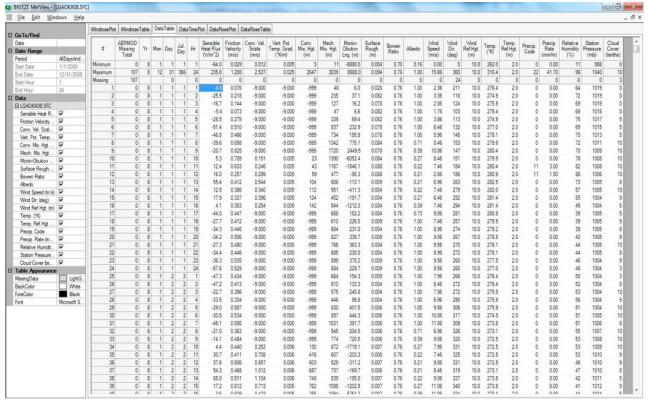
Nikolay Oskolkov, Group Leader (PI) at NIRI



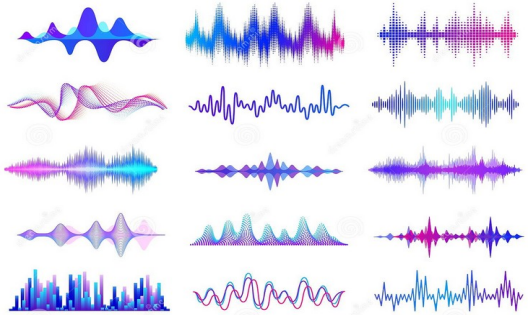
https://en.wikipedia.org/wiki/The_Emperor's_New_Clothes

Various types of data around us

Tabular



Sound



Image

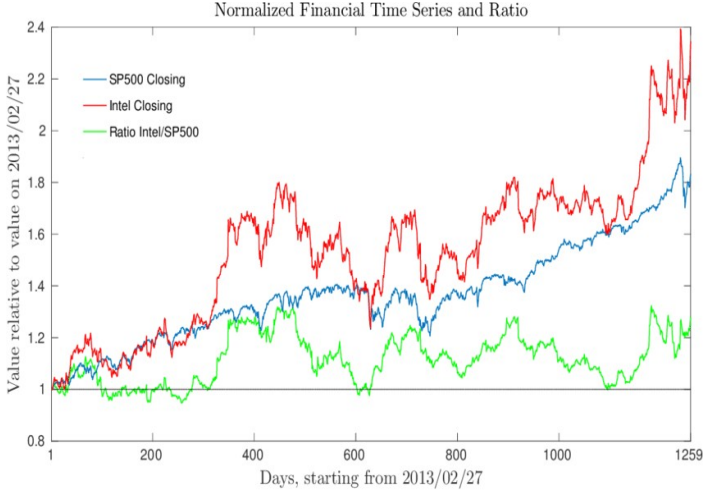


Text

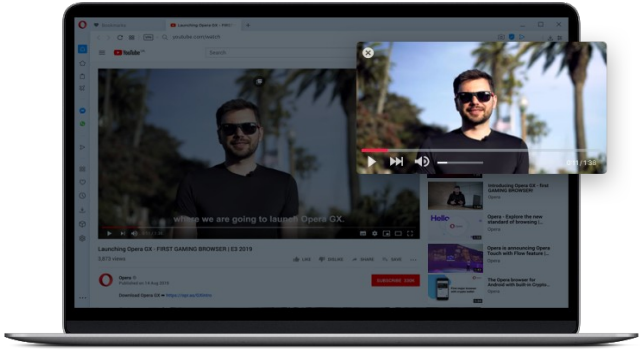


DATA

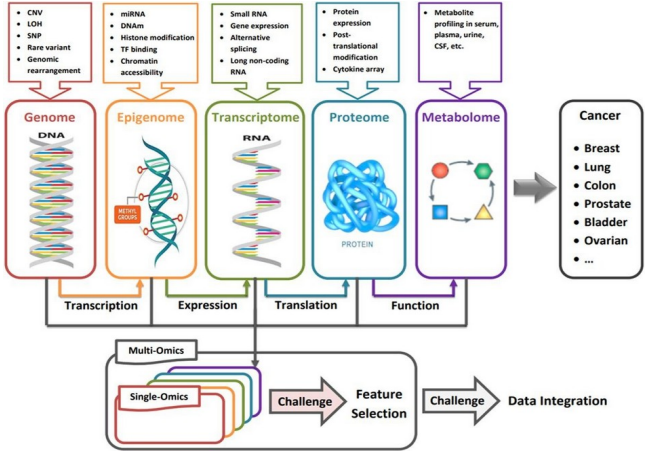
Time Series



Video



Omics Data



Genomics data are high dimensional

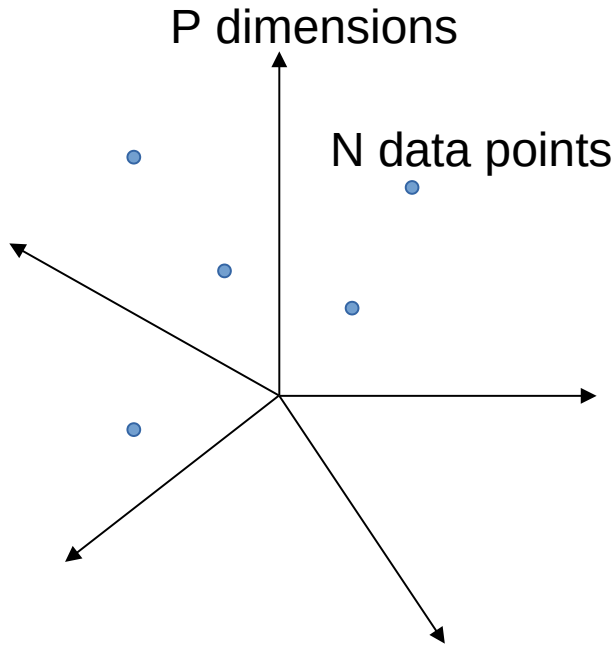
Statistical observations:
e.g. samples, cells etc.

N

P

$$\begin{pmatrix} 0 & 3 & 1 & 0 & 2 & 3 & 8 & 1 & 1 & 3 \\ 1 & 1 & 0 & 0 & 7 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 2 & 0 & 0 & 6 & 7 & 1 & 2 & 2 \\ 1 & 2 & 3 & 10 & 0 & 4 & 6 & 1 & 0 & 5 \\ 3 & 2 & 2 & 1 & 4 & 3 & 2 & 1 & 6 & 0 \\ 7 & 4 & 4 & 5 & 3 & 9 & 6 & 1 & 6 & 1 \\ 7 & 1 & 1 & 5 & 2 & 8 & 9 & 1 & 3 & 6 \\ 5 & 0 & 1 & 6 & 2 & 0 & 0 & 0 & 1 & 5 \\ 1 & 6 & 3 & 3 & 4 & 6 & 2 & 0 & 1 & 1 \\ 1 & 2 & 2 & 4 & 1 & 1 & 3 & 0 & 8 & 2 \end{pmatrix}$$

Features: genes, proteins, microbes, metabolites etc.



High Dimensional Data:
P >> N

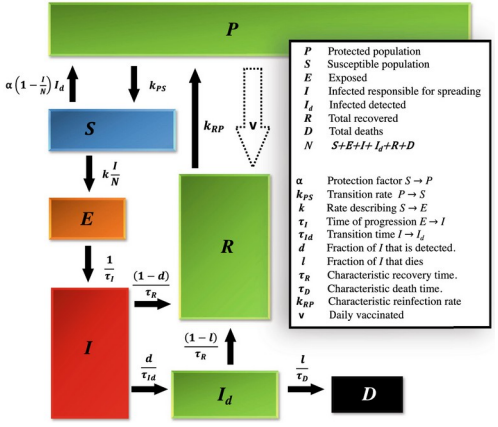
For a robust statistical analysis, one should properly “sample” the P-dimensional space, hence large sample size is required, N >> P

Some types of data analysis in Life Sciences

P is the number of features (genes, proteins, genetic variants etc.)
N is the number of observations (samples, cells, nucleotides etc.)

Biology / Biomedicine

Mathematical modeling



N → 0

Bayesianism



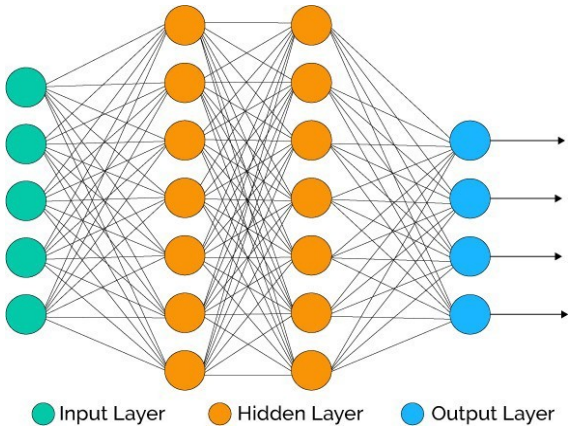
N << P

Frequentism



N ≈ P

Machine Learning



N >> P

Hypothesis-driven — **Amount of Data** — **Data-driven**

The Curse of Dimensionality

Ex.1

$$Y = \alpha + \beta X$$

$$\beta = \left(X^T X\right)^{-1} X^T Y$$

$$\left(X^T X\right)^{-1} \sim \frac{1}{\det\left(X^T X\right)} \cdots \to \infty, \quad \text{n} << \text{p}$$

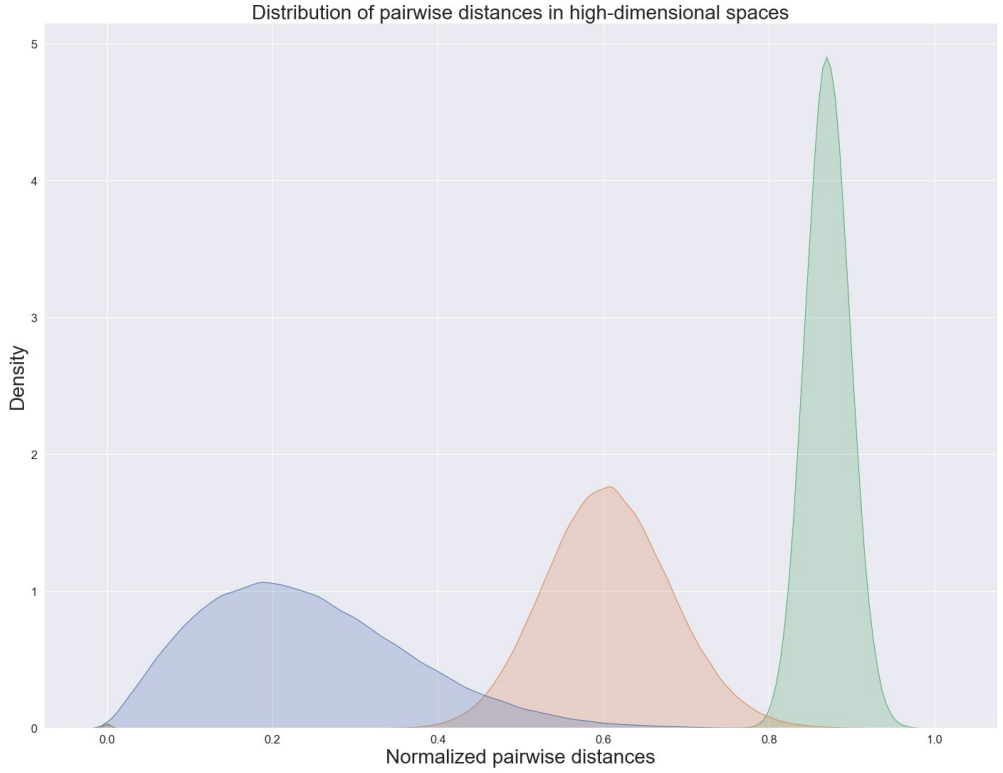
Ex.2

$$E[\hat{\sigma}^2] = \frac{n-p}{n}\sigma^2$$

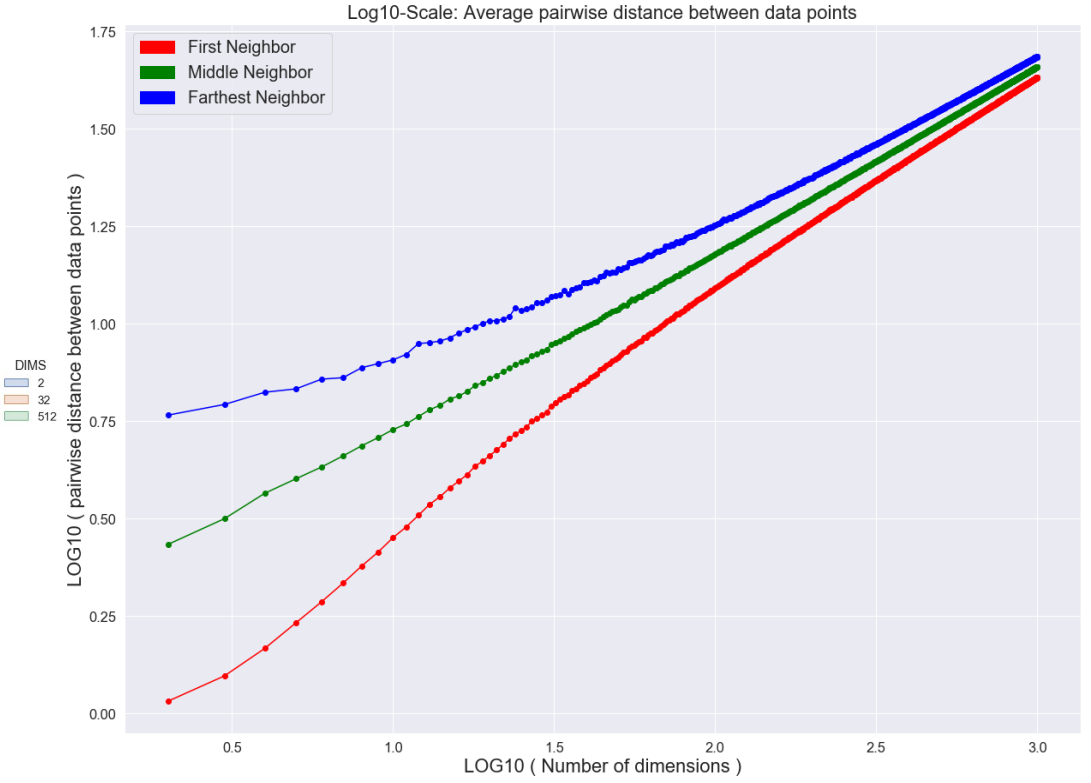
Biased ML variance estimator in HD-space

Theoretical Background of the Curse of Dimensionality

Distances in High Dimensions



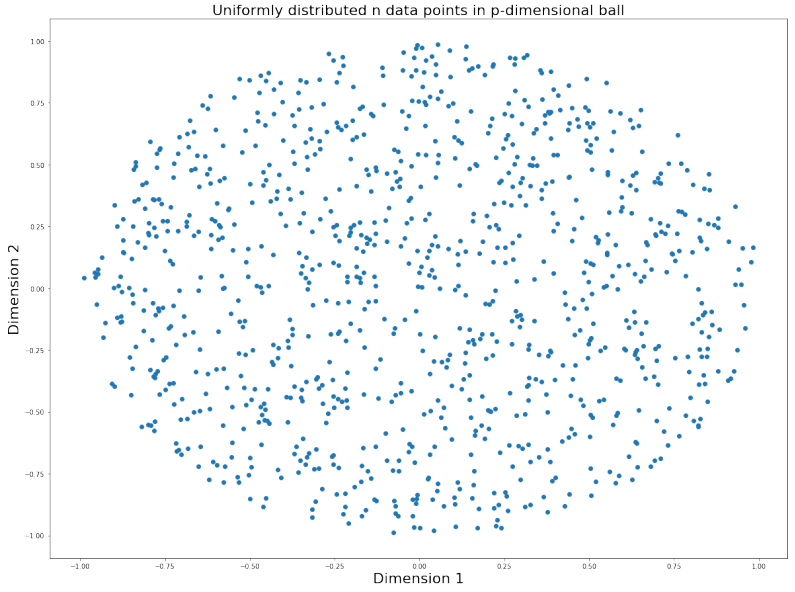
Data points become far from each other and equidistant in high dimensions

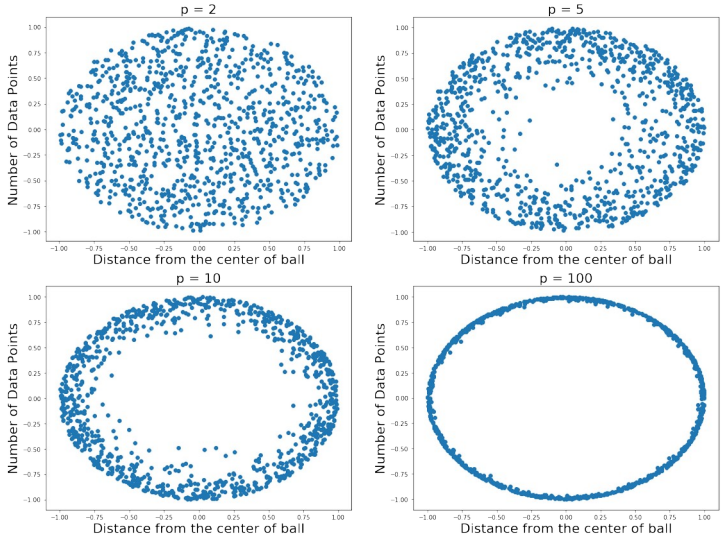


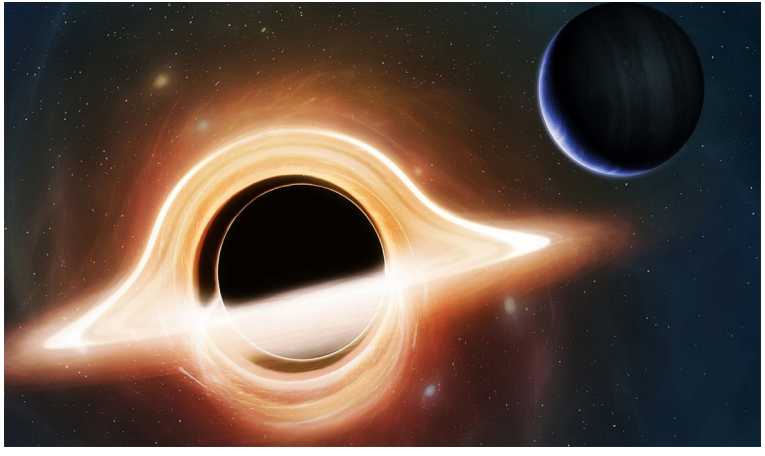
The differences between closest and farthest data point neighbours disappears in high-dimensional spaces: can't run cluster analysis

In high-dimensional space we can not separate cases and controls any more

High-dimensional data make black holes







High-dimensional data can be viewed as having a **“hole in the middle”**, hence the concept of mean / centroid loses its validity, hence we can’t use Gaussian distribution

Literature on the Curse of Dimensionality

POINTS OF SIGNIFICANCE

The curse(s) of dimensionality

There is such a thing as too much of a good thing.

Naomi Altman and Martin Krzywinski

We generally think that more information is better than less. However, in the 'big data' era, the sheer number of variables that can be collected from a single sample can be problematic. This embarrassment of riches is called the 'curse of dimensionality'[1] (CoD) and manifests itself in a variety of ways. This month, we discuss four important problems of dimensionality as it applies to data sparsity[1,2], multicollinearity[3], multiple testing[4] and overfitting[5]. These effects are amplified by poor data quality, which may increase with the number of variables.

Throughout, we use n to indicate the sample size from the population of interest and p to indicate the number of observed variables, some of which may have missing values for some samples. For example, we may have $n = 1{,}000$ subjects and $p = 200{,}000$ single-nucleotide polymorphisms (SNPs).

First, as the dimensionality p increases, the 'volume' that the samples may occupy grows rapidly. We can think of each of the n

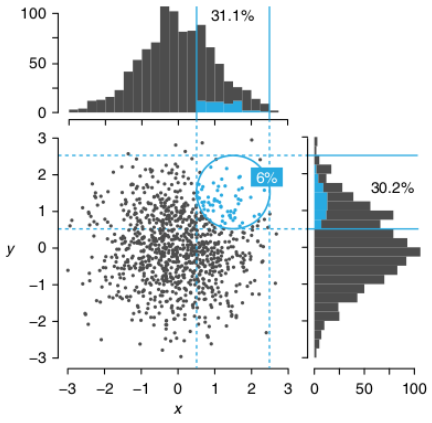


Fig. 1 | Data tend to be sparse in higher dimensions. Among 1,000 (x, y) points in which both x and y are normally distributed with a mean of 0 and s.d. $\sigma = 1$, only 6% fall within σ of $(x, y) = (1.5, 1.5)$ (blue circle). However, when the data are projected into a lower dimension—shown by histograms—about 30% of the points (all bins within blue solid lines) are within σ of 1.5. Blue

A and 100 to have the minor allele a. If we tabulate on two SNPs, A and B, we will expect only ten samples to exhibit both minor alleles with genotype ab. With SNPs A, B and C, we expect only one sample to have genotype abc, and with four or more SNPs, we expect empty cells in our table. We need a much larger sample size to observe samples with all the possible genotypes. As p increases, we may quickly find that there are no samples with similar values of a predictor.

Even with just five SNPs, our ability to predict and classify the samples is impeded because of the small number of subjects that have similar genotypes. In situations where there are many gene variants, this effect is exacerbated, and it may be very difficult to find affected subjects with similar genotypes and hence to predict or classify on the basis of genetic similarity.

If we treat the distance between points (e.g., Euclidian distance) as a measure of similarity, then we interpret greater distance as greater dissimilarity. As p increases, this

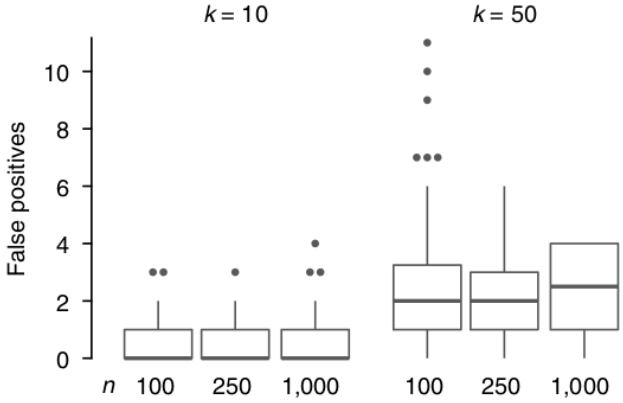


Fig. 3 | The number of false positives increases with each additional predictor. The box plots show the number of false positive regression-fit P values (tested at $\alpha = 0.05$) of 100 simulated multiple regression fits on various numbers of samples ($n = 100$, 250 and 1,000) in the presence of one true predictor and $k = 10$ and 50 extraneous uncorrelated predictors. Box plots show means (black center lines), 25th and 75th percentiles (box edges), and minimum and maximum values (whiskers). Outliers (dots) are jittered.

Altman N, Krzywinski M. The curse(s) of dimensionality. Nat Methods. 2018 Jun;15(6):399-400. doi: 10.1038/s41592-018-0019-x. PMID: 29855577.

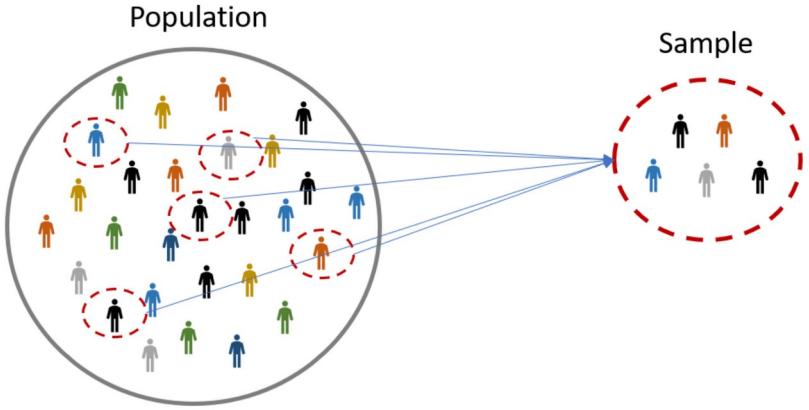
Correcting for multiple testing does not solve the problem of too many false-positive hits

How to Combat the Curse of Dimensionality?



Regularization

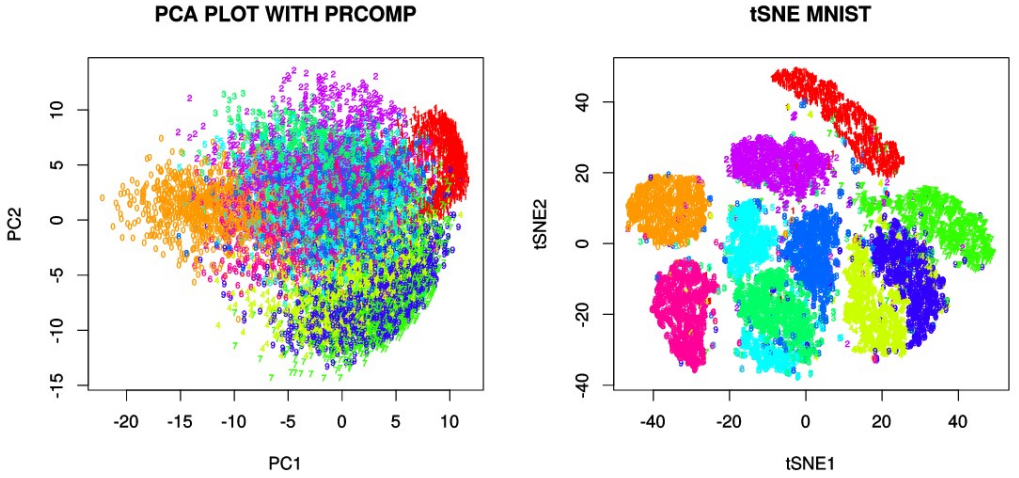
Increase sample size: epidemiological limit





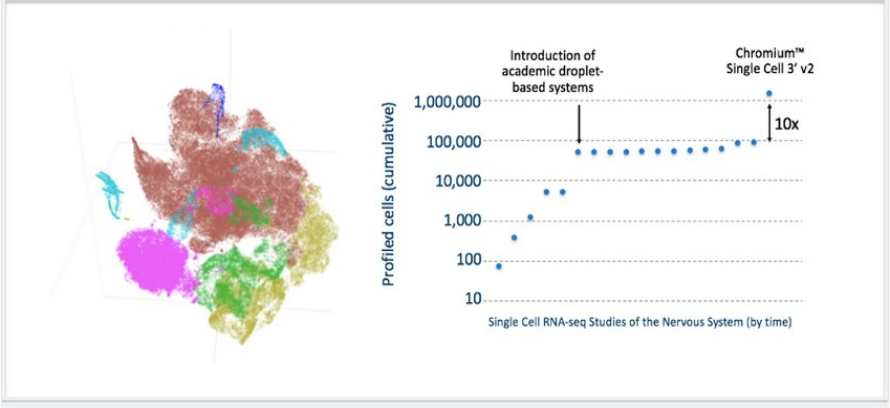
Bayesian statistics

Dimension Reduction



Big Data in Single Cell Genomics

« Back to Blog ‹ Newer Article Older Article ›



Our 1.3 million single cell dataset is ready to download

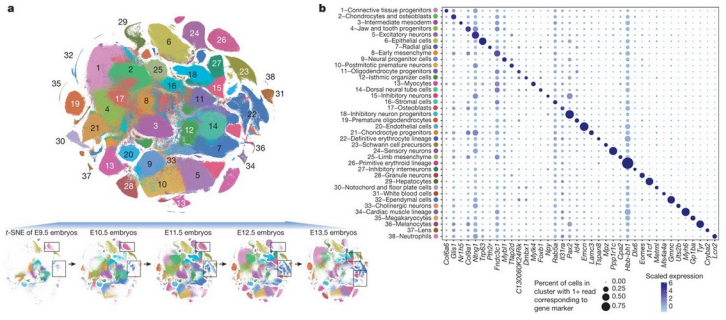
0 KUDOS

POSTED BY: grace-10x, on Feb 21, 2017 at 2:28 PM

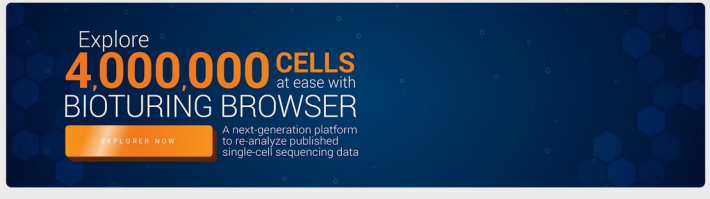
At ASHG last year, we announced our 1.3 Million Brain Cell Dataset, which is, to date, the largest dataset published in the single cell RNA-sequencing (scRNA-seq) field. Using the Chromium™ Single Cell 3′ Solution (v2 Chemistry), we were able to sequence and profile 1,308,421 individual cells from embryonic mice brains. Read more in our application note Transcriptional Profiling of 1.3 Million Brain Cells with the Chromium™ Single Cell 3′ Solution.

Fig. 2: Identifying the major cell types of mouse organogenesis.

From: The single-cell transcriptional landscape of mammalian organogenesis



a, t-SNE visualization of 2,026,641 mouse embryo cells (after removing a putative doublet cluster), coloured by cluster identity (ID) from Louvain clustering (in **b**), and annotated on the basis of marker genes. The same t-SNE is plotted below, showing only cells from each stage (cell numbers from left to right: n = 151,000 for E9.5; 370,279 for E10.5; 602,784 for E11.5; 468,088 for E12.5; 434,490 for E13.5). Primitive erythroid (transient) and definitive erythroid (expanding) clusters are boxed. **b**, Dot plot showing expression of one selected marker gene per cell type. The size of the dot encodes the percentage of cells within a cell type in



Single Cell Analysis

5,500,000 cells will be indexed into BioTuring Single-cell Data Repository this September

by biomembers • August 30, 2019

Human Cell Atlas, single-cell data

We are glad to announce that we will upsize the current single-cell database in BioTuring Single-cell Browser to 5,500,000 cells this September. With this release, we will double the current number of publications indexed in BioTuring Single-cell Browser, and cross the number of cells hosted on available public single-cell data repositories like Human Cell Atlas (HCA) and Broad Institute's Single-cell Portal.

Search

RECENT POSTS

A new tool to interactively visualize single-cell objects (Seurat, Scanpy, SingleCellExperiments, ...)
September 26, 2019

5,500,000 cells will be indexed into BioTuring Single-cell Data Repository this September
August 30, 2019

Typical Dimensions of Omics Data

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Metabolomics
N ≈ P

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Proteomics
N ≈ P

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Metagenomics
N ≈ P

← **manageable**

P

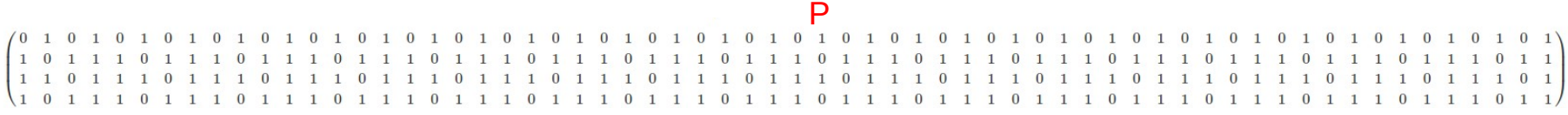
N $\begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{pmatrix}$

Transcriptomics
N << P
(Single cell: N <= P)

challenging

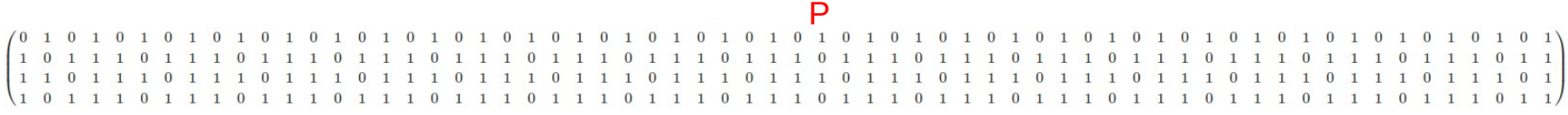
↓

P

N 

Genomics
N <<<< P

P

N 

Methylomics
N <<<< P

Low Dimensional Space

```r
set.seed(123); n <- 20; p <- 2
Y <- rnorm(n); X <- matrix(rnorm(n*p),n,p); summary(lm(Y~X))
```

```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.0522 -0.6380  0.1451  0.3911  1.8829
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.14950    0.22949   0.651    0.523
## X1          -0.09405    0.28245  -0.333    0.743
## X2          -0.11919    0.24486  -0.487    0.633
##
## Residual standard error: 1.017 on 17 degrees of freedom
## Multiple R-squared:  0.02204,    Adjusted R-squared:  -0.09301
## F-statistic: 0.1916 on 2 and 17 DF,  p-value: 0.8274
```

Going to Higher Dimensions

```
set.seed(123456); n <- 20; p <- 10
Y <- rnorm(n); X <- matrix(rnorm(n*p),n,p); summary(lm(Y~X))
```

```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0255 -0.4320  0.1056  0.4493  1.0617
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.54916    0.26472   2.075   0.0679 .
## X1           0.30013    0.21690   1.384   0.1998
## X2           0.68053    0.27693   2.457   0.0363 *
## X3          -0.10675    0.26010  -0.410   0.6911
## X4          -0.21367    0.33690  -0.634   0.5417
## X5          -0.19123    0.31881  -0.600   0.5634
## X6           0.81074    0.25221   3.214   0.0106 *
## X7           0.09634    0.24143   0.399   0.6992
## X8          -0.29864    0.19004  -1.571   0.1505
## X9          -0.78175    0.35408  -2.208   0.0546 .
## X10          0.83736    0.36936   2.267   0.0496 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8692 on 9 degrees of freedom
## Multiple R-squared:  0.6592,    Adjusted R-squared:  0.2805
```

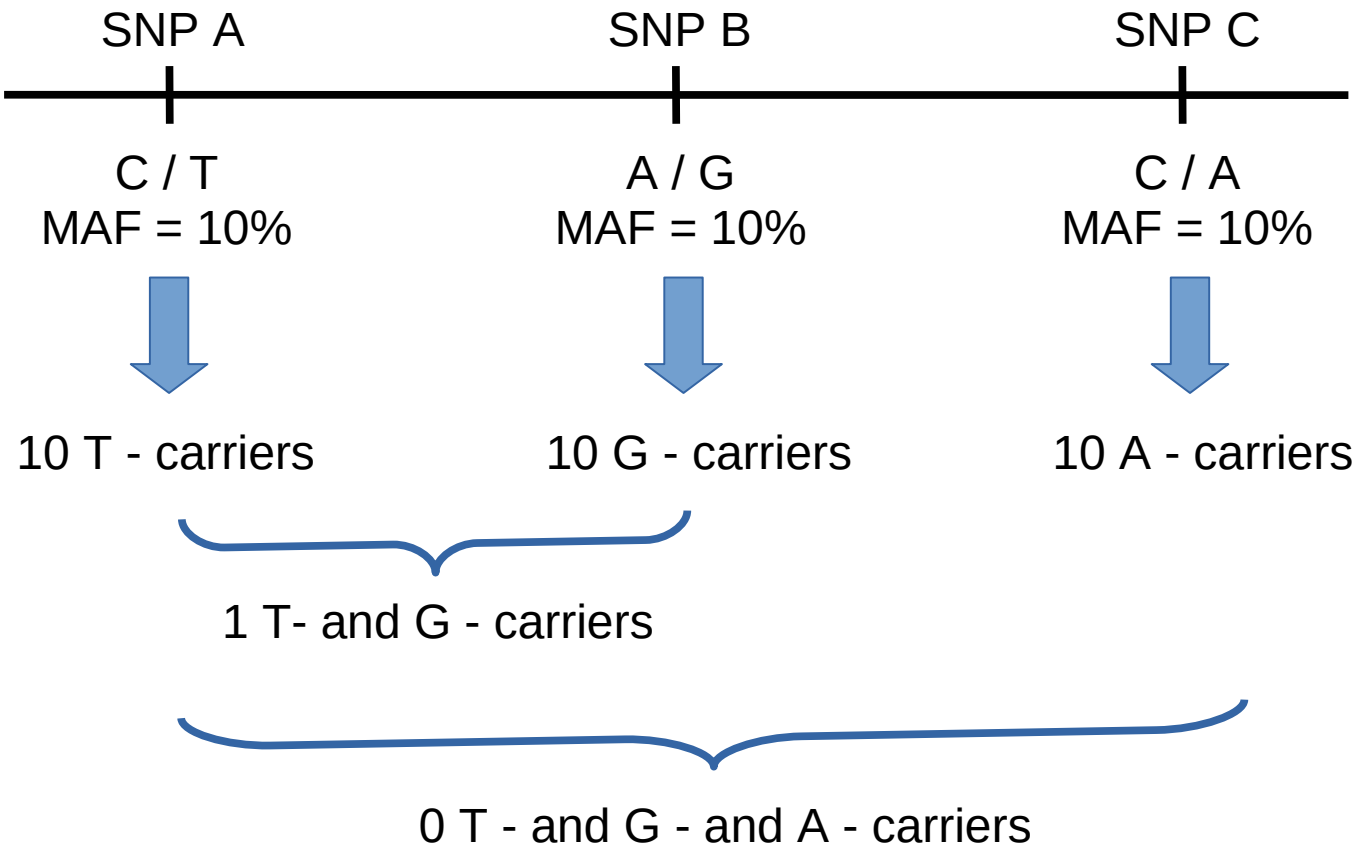
Even Higher Dimensions

```
set.seed(123456); n <- 20; p <- 20
Y <- rnorm(n); X <- matrix(rnorm(n*p),n,p); summary(lm(Y~X))
```

```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
## ALL 20 residuals are 0: no residual degrees of freedom!
##
## Coefficients: (1 not defined because of singularities)
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.34889         NA      NA       NA
## X1           0.66218         NA      NA       NA
## X2           0.76212         NA      NA       NA
## X3          -1.35033         NA      NA       NA
## X4          -0.57487         NA      NA       NA
## X5           0.02142         NA      NA       NA
## X6           0.40290         NA      NA       NA
## X7           0.03313         NA      NA       NA
## X8          -0.31983         NA      NA       NA
## X9          -0.92833         NA      NA       NA
## X10          0.18091         NA      NA       NA
## X11         -1.37618         NA      NA       NA
## X12          2.11438         NA      NA       NA
## X13         -1.75103         NA      NA       NA
## X14         -1.55073         NA      NA       NA
## X15          0.01112         NA      NA       NA
```

How The Curse of Dimensionality Affects Genomics Research

Sparsity of Genomics Data



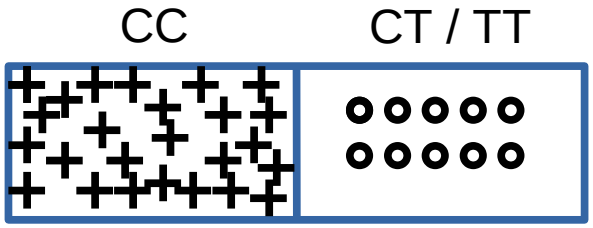
Sparsity of Genomics Data

N = 100

SNP A (C / T, MAF = 10%) and SNP B (A / G, MAF = 10%)

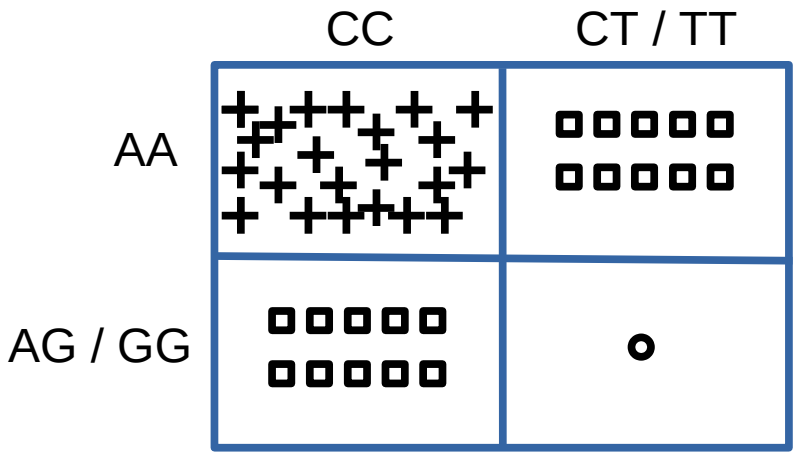
Phen ~ SNP A + SNP B

1D SNP A:



- minor T – allele carrier

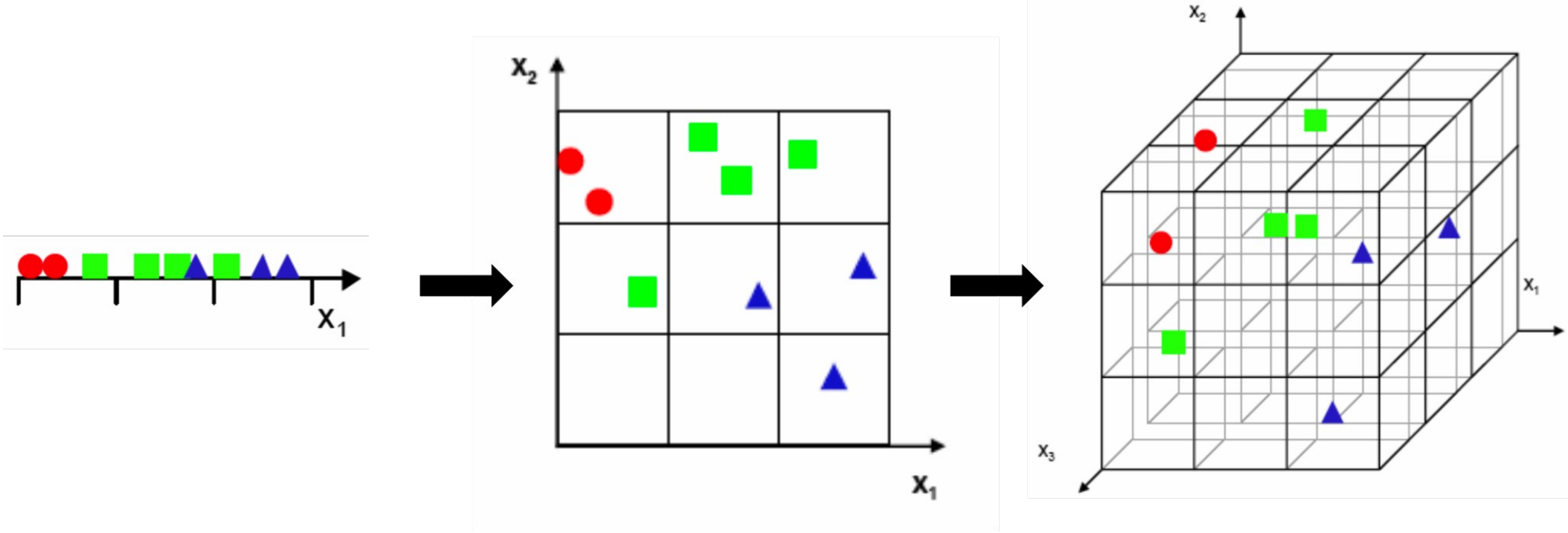
2D SNP A
+
SNP B:



- carrier of both minor T - and G - alleles

- carrier of either minor T - or G - alleles

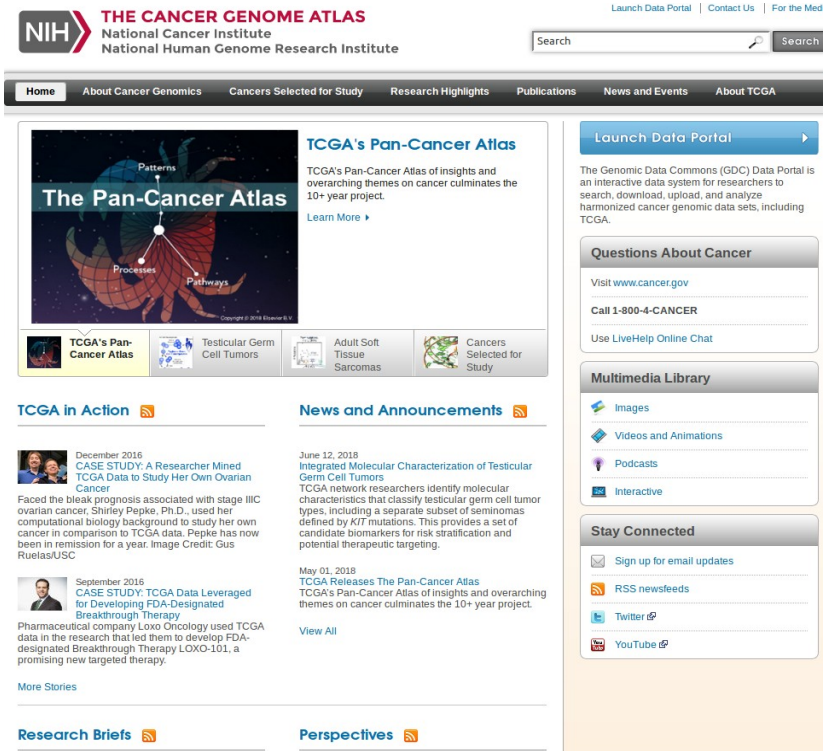
Data Sparsity in High Dimensions

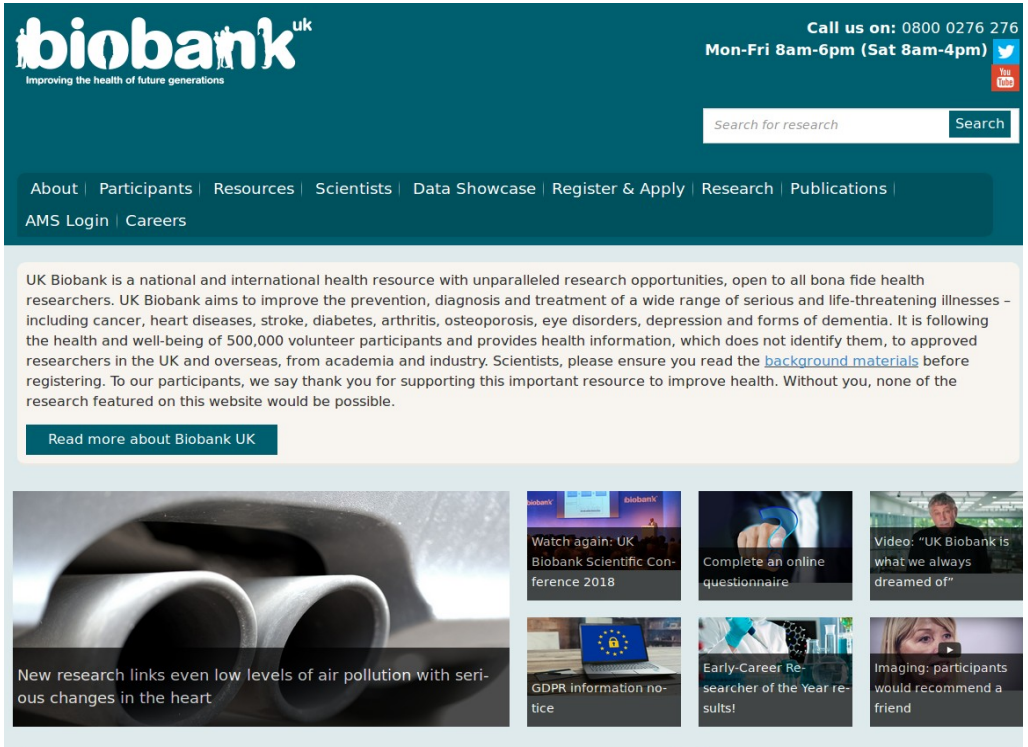


Volume of high-dimensional data grows exponentially with the number of dimensions

A lot of free space appears, very difficult to sample (have observations for the whole range of values on each dimension / axis) that large space

Do We Have Big Data in Biomedicine?



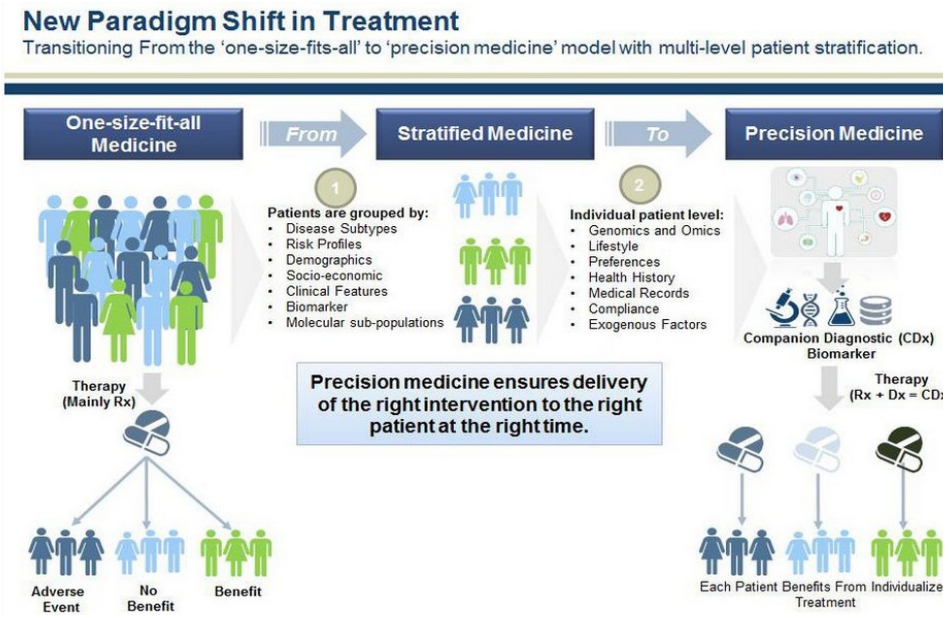




Mark McCarthy:

Now we have found over 400 genetic signals influencing type 2 diabetes, what on earth are we going to do with them?

Precision Medicine: Why isn't it in the Clinics?





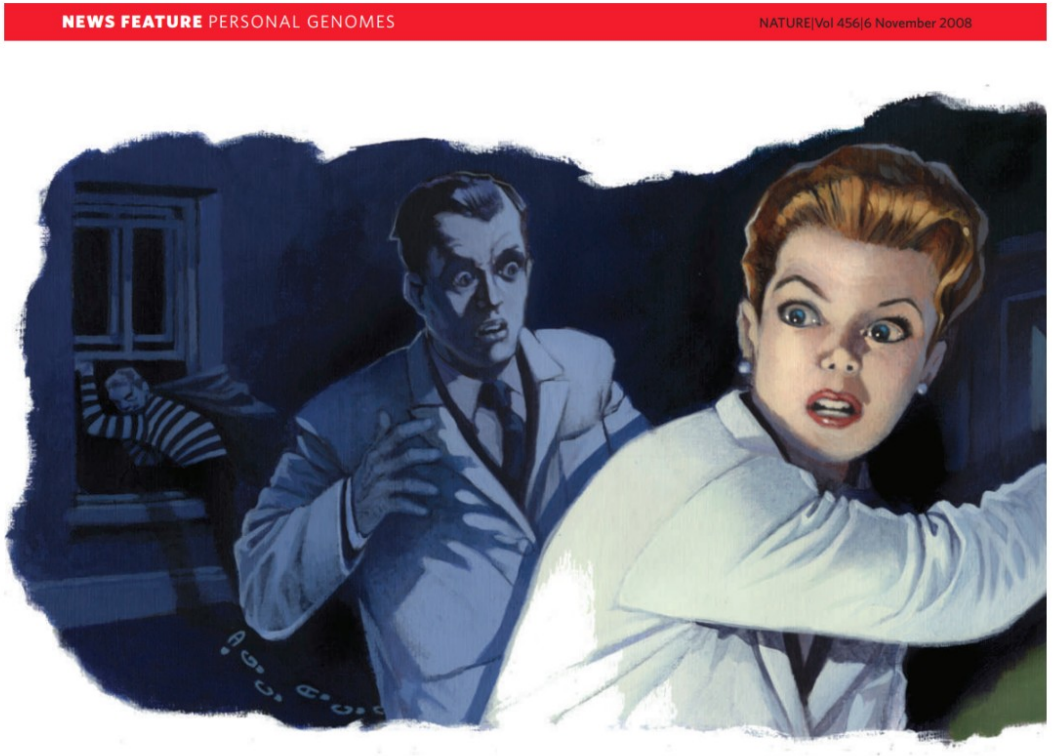
Typical debate at Precision Medicine conferences:

1) Leaders: “We sequence a lot, it is important”
2) Clinicians: “This does not help us in the Clinics”

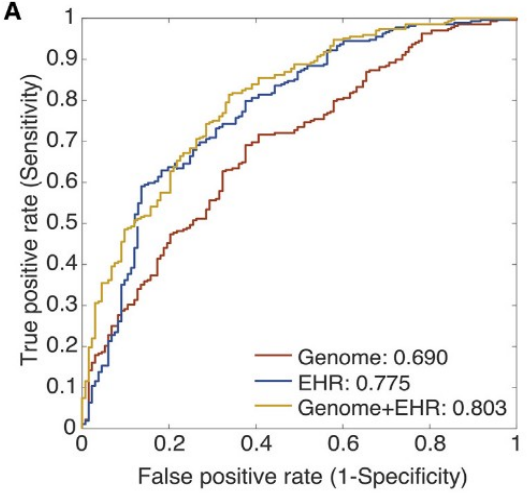
The Missing Heritability Problem

GRS = beta1 * SNP1 + beta2 * SNP2 + … ← training

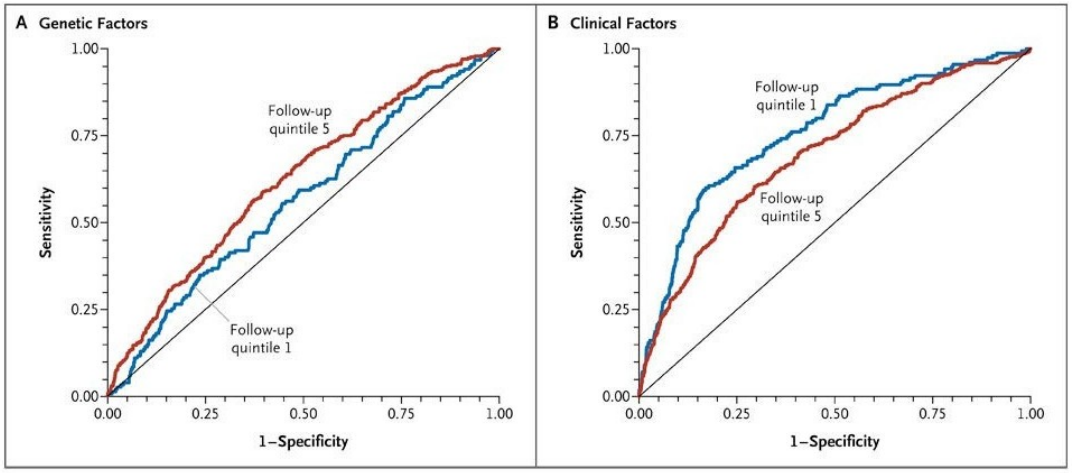
Phenotype ~ GRS => Adjusted R^2 ← validation



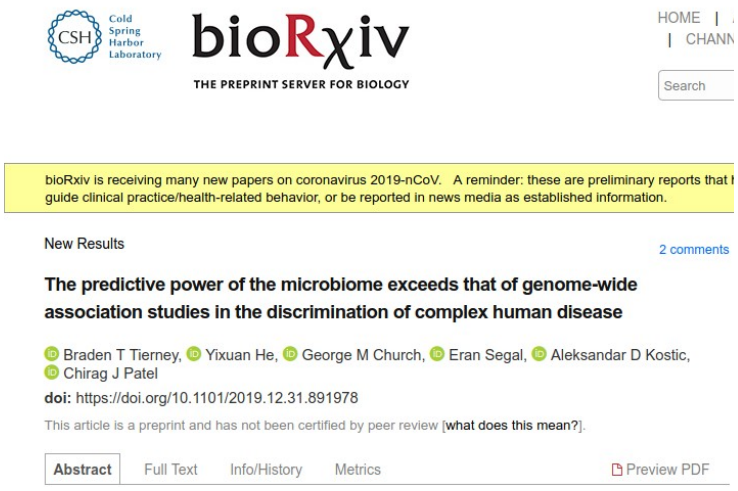
Precision Medicine: Poor Genomics Prediction

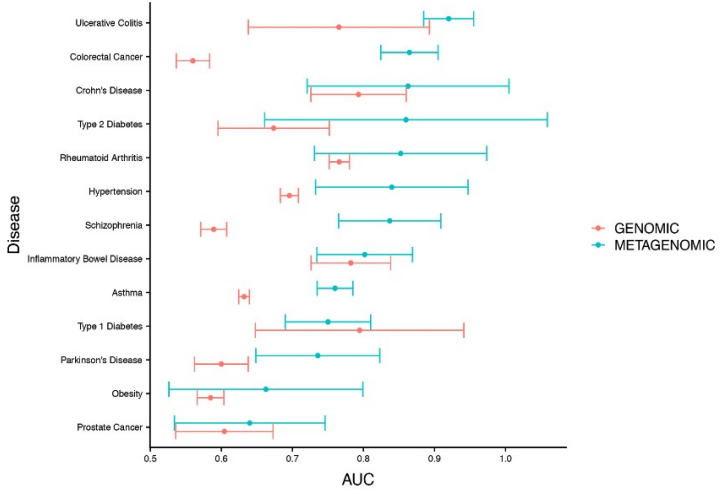


Li et al., 2018, Cell 174, 1361-1372
Decoding the Genomics of Abdominal Aortic Aneurism



Lyssenko et al., 2008, New England Journal of Medicine 359, 2220-2232
Clinical Risk Factors, DNA Variants and Development of Type 2 Diabetes





Human Height Missing Heritability Case Solved

NEWS · 23 APRIL 2019

Genetic study homes in on height's heritability mystery

Analysis of 20,000 genomes reveals that rare gene variants can help to explain how the trait is inherited.

Linda Geddes



Height is often inherited — but the precise genetics factors have long eluded researchers. Credit: Shomos Uddin/Getty

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New Results

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Recovery of trait heritability from whole genome sequence data

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TOPMed Anthropometry Working Group, Trans-Omics for Precision Medicine Consortium,
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doi: https://doi.org/10.1101/588020

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Abstract

Heritability, the proportion of phenotypic variance explained by genetic factors, can be estimated from pedigree data [1], but such estimates are uninformative with respect to the underlying genetic architecture. Analyses of data from genome-wide association studies (GWAS) on unrelated individuals have shown that for human traits and disease, approximately one-third to two-thirds of heritability is captured by common SNPs [2–5]. It is not known whether the remaining heritability is due to the imperfect tagging of causal variants by common SNPs, in particular if the causal variants are rare, or other reasons such as over-estimation of heritability from pedigree data. Here we show that pedigree heritability for height and body mass index (BMI) appears to be fully recovered from whole-genome sequence (WGS) data on 21,620 unrelated individuals of European ancestry. We assigned 47.1 million genetic variants

Posted March 25, 2019.

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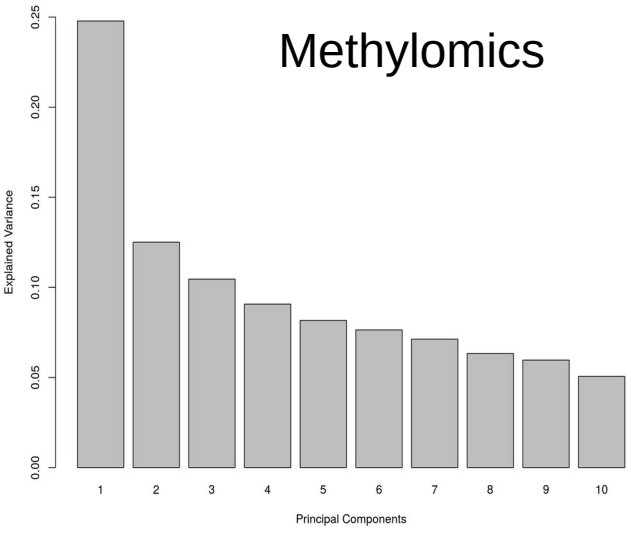
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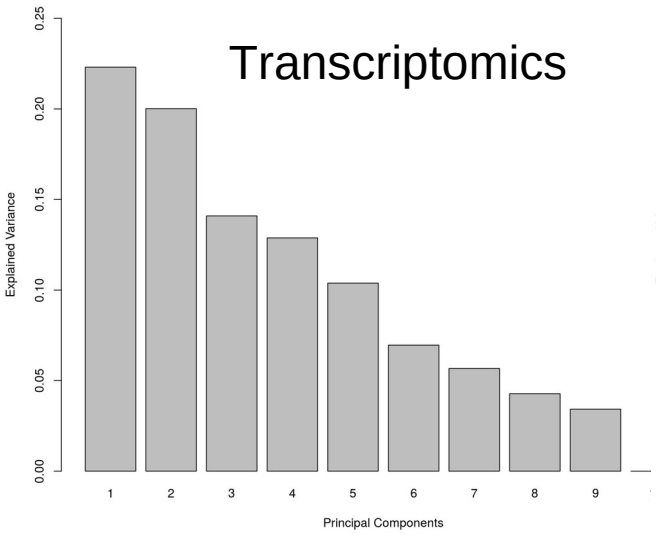
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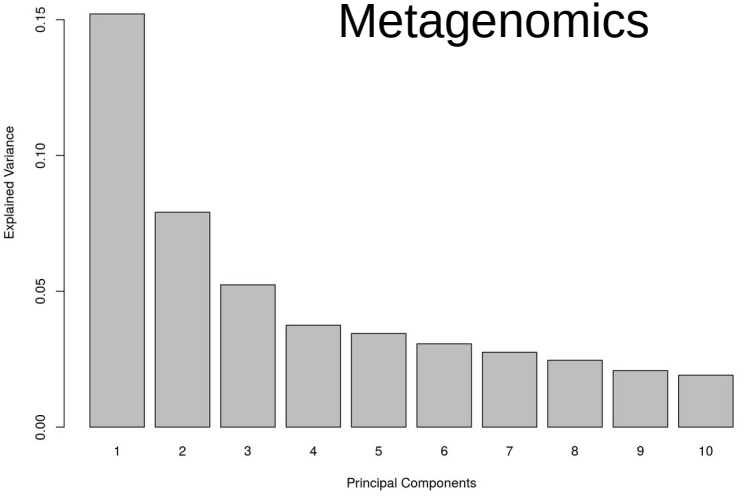
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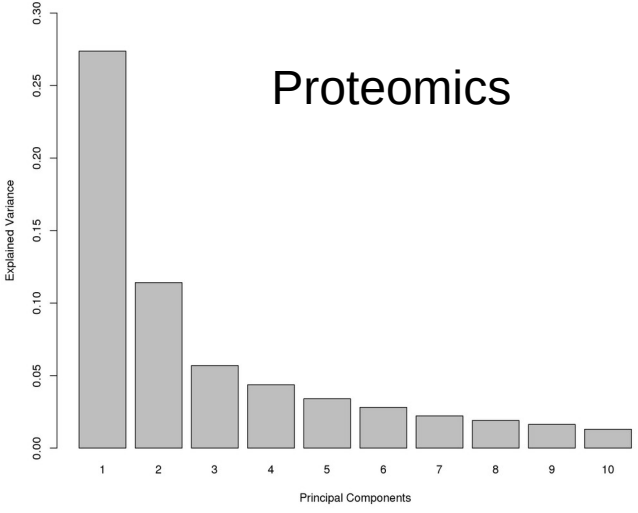
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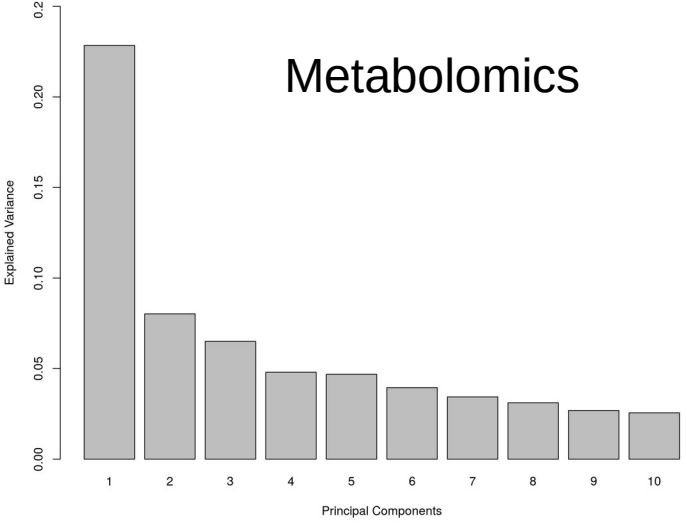
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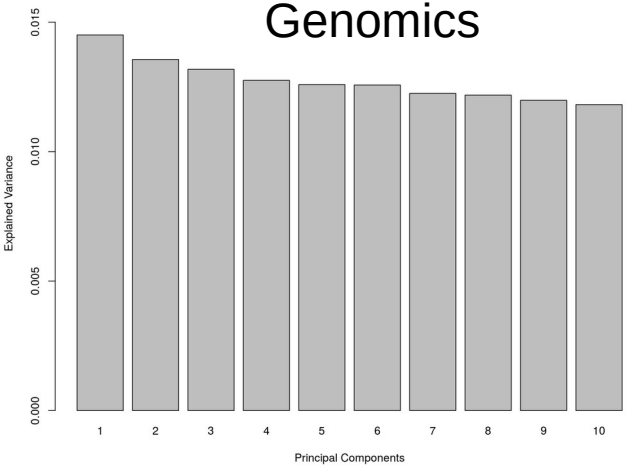
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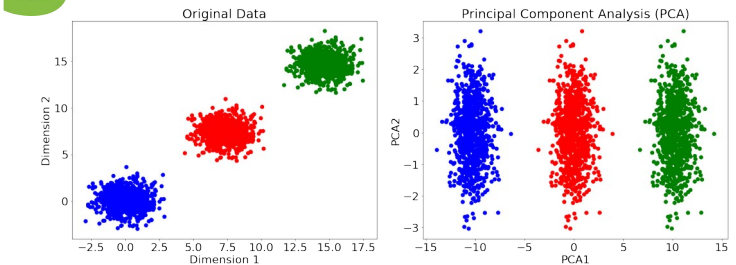
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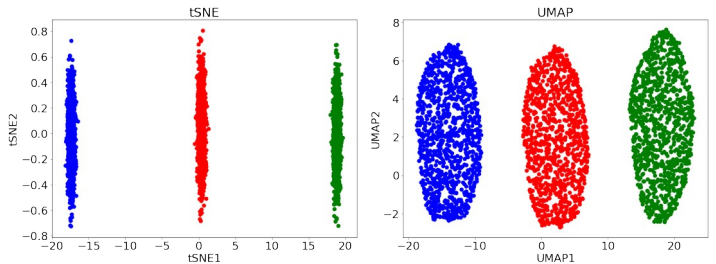


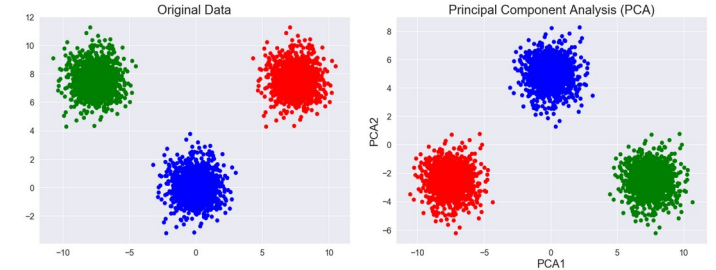
Genomics

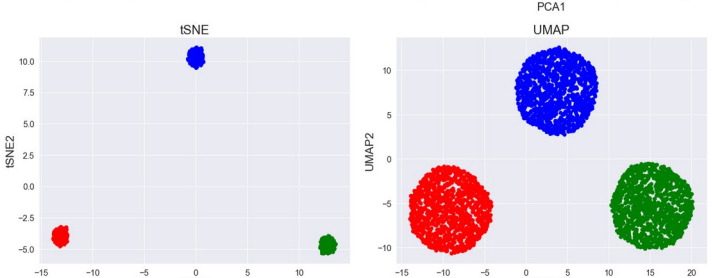


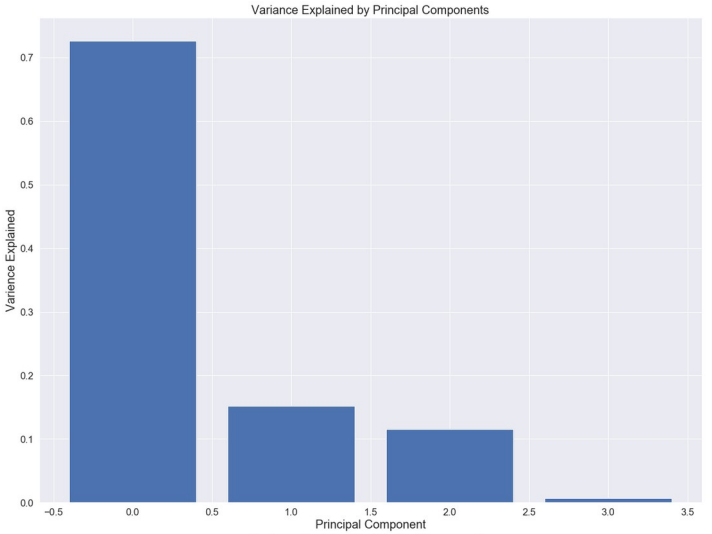
Symmetry of scRNAseq vs Genetic Variation



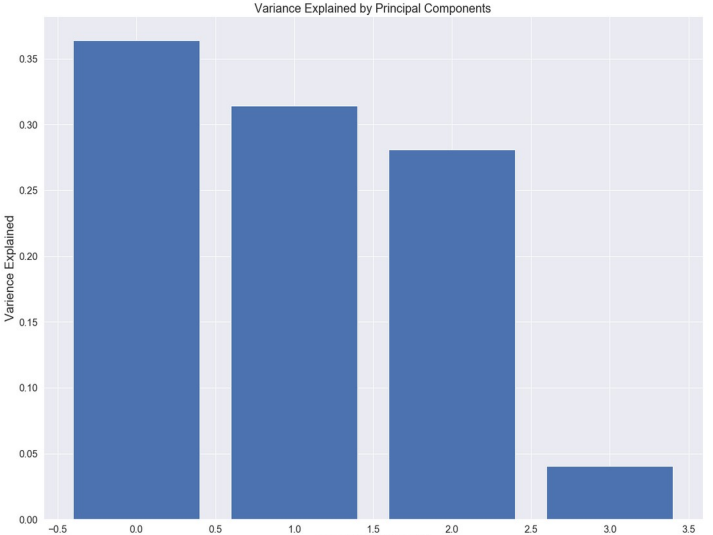








scRNAseq



Genetic Variation

National Bioinformatics Infrastricture Sweden (NBIS)



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